

Homework 2

Deadline for Submission: 15th June, 2026

Objectives

1. This assignment focuses on interactivity of the visualization
2. Visualization of a high-dimensional dataset through several interaction mechanisms

Dataset: A version of the country factbook dataset

This is a tabular dataset that compiles a number of interesting attributes about many countries:

- Area: Total surface area in square kilometers.
- Country: Name of country
- population: Current total population count.
- Total fertility rate:
- birth_rate / death_rate: Annual births and deaths per 1,000 people.
- life_exp_at_birth: Total, male, and female life expectancy in years.
- Infant mortality rate
- Current account balance:
- gdp: Gross Domestic Product
- gdp per capita: Average economic output per individual
- GDP real growth rate:
- inflation_rate: The annual percentage change in consumer prices.
- Labor force
- unemployment_rate: Percent of the active labor force without employment.
- Industrial production growth rate:
- Exports/Imports: in USD
- Investment
- Public debt
- Military expenditures
- Reserves of foreign exchange & gold
- electricity_production / electricity_consumption: Total annual power generated and used in kilowatt-hours.
- Natural gas consumption
- Oil consumption
- internet_users: Absolute count of the population with internet access.
- highways/railways: Total road and track lengths measured in kilometers

Task 1: Bubble chart

A bubble chart is an extension of a scatter plot by varying the size of the points (turning them into "bubbles") to represent a third variable.

Implement a basic bubble chart visualization with individual data points drawn as circles whose area and color vary as a function of two variables (in addition to the two variables that control the horizontal and vertical position of each circle). The horizontal (x) coordinate corresponds to 'GDP per capita', the vertical (y) coordinate corresponds to 'Life expectancy at birth', the area corresponds to 'Population', and the color corresponds to 'Birth rate'. Make sure to select scales, ranges, and color maps appropriately.

Deliverable: p2_bubbles.py

Homework 2

Task 2: Widgets

In Task 1, you were asked to assign specific (and arbitrary) attributes to each degree of freedom of your bubble chart. In reality, the visualization would be much more useful if the user could decide at runtime what attributes should be displayed and how. The goal of Task 2 is to create a user interface that makes this dynamic selection possible. Practically, you will augment the visualization of the Task 1 with widgets that collectively allow one to select for each of x, y, radius, and color, what attribute of the dataset should be used. To do so, your widget should display a list of the existing attributes. Here, the selection of each displayed attribute should be made through a drop down menu. An example integrating a drop down menu in a Bokeh program is shown at https://docs.bokeh.org/en/latest/docs/user_guide/interaction/widgets.html#dropdown.

A second aspect of the visualization that the user might want to control interactively is the scaling of the area with respect to the selected attribute. To enable this, you will add a slider widget to your visualization, whose value selection will be tied to the area of the circles in your bubble chart visualization. An example of slider bars associated with a Bokeh visualization is demonstrated at https://docs.bokeh.org/en/latest/docs/user_guide/interaction/widgets.html#rangeslider.

Deliverable: p2_widgets.py

Task 3: Linked Brushing

We saw in Homework 1 that a matrix of scatter plots (plotting all attributes vs. one another) offers a comprehensive view of the data through a set of a 2D projections. For this third task, we will follow a similar approach and display four bubble charts in a 2-by-2 array to visualize more attributes than what is possible with a single bubble chart. The user should be free to select different attributes to control the appearance of each chart. For that, you will need to quadruple the number of the widgets created in the previous task.

One issue that you will face with this visualization is to understand how data points and patterns present in each chart relate to each other across charts. One way to mitigate this problem consists in selecting the same attributes to control the radius and/or color of the bubbles in all charts. An alternative, and more flexible, solution is to use what is known as linked brushing – you should investigate linked brushing at

https://docs.bokeh.org/en/latest/docs/user_guide/interaction/linking.html#linked-brushing. Using linked brushing, the local selection of some data points in a visualization triggers the selection of the same data points in another visualization. You should implement this method for this third task and make sure that a selection made in any of the four windows is replicated in the other three. Practically, your implementation should allow for data points to be selected within a rectangular region drawn by the user and this selection should be visually accompanied by a change in the appearance of the selected items across all four windows to make them easy to spot.

Deliverable: p2_brushing.py

Task 4: Details on Demand

Though four bubble charts allow one to display a substantial number of attributes at once, they can only convey a relatively small portion of the information available in this dataset. A typical means to overcome this limitation is to give the user the option to query additional details about the items included in the visualization.

Homework 2

To afford access to additional details, you will use the hover event mechanism to trigger the display of all the attributes associated with a given data point in a tooltip. Specifically, upon moving the mouse over a particular data point, a window should appear to the side of the corresponding location that displays the name of the country along with a series of (attribute name, value) pairs that show a subset of the information available for this country. In addition, the country selected should be highlighted across all four charts. Your fourth task is to add this feature to the visualization tool that you created in Task 3. As a starting point for this task, you should find helpful to refer to the following:

https://docs.bokeh.org/en/latest/docs/user_guide/interaction/tooltips.html and
<https://docs.bokeh.org/en/latest/docs/examples/plotting/hover.html>.

Deliverable: p2_tooltip.py